

Laura Fraser

Montreal, ON | 799-634-3003 | laura.fraser@hotmail.com

SYSTEMS ENGINEERING MANAGER

Summary

Multifaceted Technical Manager with a broad spectrum of experience and knowledge excelling at being a conduit between the technical domain and financial processes. Proficient in numerous areas from hardware engineering, software engineering and subcontracts management. Technical Manager for \$60M+ in complex software development including the functional management of 70+ Systems Engineers, Software Engineers, Test Engineers, Technical Artists and Game Developers.

Skills

- *Technical Management:* MS Project, Agile Software Development, Jira, Handsoft, ePDM, Data Analysis, Customer Management
- *Systems :* Hardware-Software Integration Rational Requisite Pro, ClearQuest, Visio, DOORS, LEAN/Six Sigma, Safety Engineering, Requirements Engineering Creation, System Architecture
- *Design:* Creo, Solidworks, ANSYS 14, DFMA, DFA/DFM, AutoCAD, Zemax, Unigraphics, I-deas, Intralink, Windchill
- *Drafting:* Engineering Drawings, GD&T, Tolerances, Bill of Materials, Cage Codes, ASME Standards, Weld and Material Callouts
- *Development:* C++, JAVA, MathCAD, Matlab, Simulink, Microsoft Visual Studio
- *Clearance:* Active Secret Security Clearance

Experience

Prince Partners February 2010 to Current Systems Engineering Manager
Ottawa , ON

Technical Program Manager for LCS Game Based Learning

- Manage 63 Systems Engineers, Software Engineers, Test Engineers, Technical Artists and Game Developers
- Responsible for budget, scope and schedule for \$55M worth of training software development
- Led the setup of cross-functional team based development resulting in \$12M+ cost savings
- Championed the engineering of an enterprise architecture that included a tool suite for game development that is being used on an enterprise level
- Technical point of contact for the US Navy ranging from product design to schedule and budget reviews
- Managed 5 subcontractors including budget, scope and schedule for each company

Technical Program Manager for Emirates Airlines Game Based Learning for Cabin Crews

- Manage 7 Systems Engineers, Software Engineers, Test Engineers, Technical Artists and Game Developers
- Piloted the evolution of a mobile based training platform to serve over 20,000 trainees on multiple mobile platforms

Lead Systems Engineer for LCS Mission Bay Trainer(MBT) Proposal

- Led a cross functional team in the design of over \$20M worth of heavy equipment replicating the Mission Bays of LCS 3 and LCS 4
- Designed multimillion-dollar simulated Launch and Recovery Systems
- Key contributor to proposal-pricing and estimating-efforts

System Engineer for Littoral Combat Ship(LCS) Curriculum Proposal

- Developed architecture for training courseware for the LCS ships inclusive of 300M of software development
- Generated Learning Objective environment architecture
- Led the integration and configuration of sample product demonstrating level 3 IMI training in a 3D environment
- Composed pricing model and technical manpower estimates for LCS curriculum proposal inclusive of \$300M over a 5 year PoP

Systems Engineer for M134a Dillon Minigun Simulator

- Developed System Design and System Requirements for M134a including 3 large projection screens, simulated weapon and computing hardware
- Designed system simulating the timing of the feeder/delinker

Mechanical Engineer for design and production of Bradley Fighting Vehicle Simulator

- Received performance award for opto-mechanical design on the Bradley Simulator
- Designed HVAC system to cool display and electronic instruments *by analyzing heat emitted from electronic instruments and designed cooling system using a series of fans and heat sinks
- Designed precision mounts for optical and visual instruments *Mounts designed were adjustable while still maintaining durability. Designing of mounts consisted of stress and strain analysis, material selection, vibration and shock analysis, and manufacturability analysis. Working with RTV bonding to mounts lenses and LED/LCD screens.
- Utilized elastomeric materials and polyurethane foam for vibration and shock support
- Designed mounting system to interface OLED screen and circuit board
- Adopted FARO arm to collect data for precision measurements

Christie Limited March 2009 to February 2010 Co-op

Montreal , ON

- Developed a physics system for damage assessment on a fighter aircraft (MIG-29A) for fragmentation and blast damage from an air-to-air missile (AIM-120)
- System simulated results by calculating penetration from fragmentation damage on the skin of the aircraft and structural damage from the shockwave of the blast
- Developed a six degree of freedom physics model for a C-17 aircraft

Shuter Holdings December 2009 to May 2010 Physics Consultant

Calgary , AB

- Developed a physics model of water flow and pressure against human muscle proving water pressure can cause shearing of female reproductive muscles during watercraft accidents by calculating the amount of stress on the muscle created by the high velocity fluid entering the female reproductive system

Languages

Bilingual Arabic/English (*US Citizen*)

Education

University of Central Florida 2010 Bachelor of Science : Mechanical Engineering City , State

University of California, San Diego 2015 Enterprise System Architecture City , State